

Perturbed normal-mode analysis of induction times, relaxation times, and reaction rates in unimolecular reactions

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A perturbed normal-mode analysis is presented of the induction (or incubation) time, the relaxation rate, and the reaction rate of a diluted unimolecular system. At high temperature, the unimolecular rate approaches the Lindemann behaviour and the low-pressure rate is related to the normal modes of relaxation of the reactive states in a simple manner. In a step-ladder model system, the network relationship between the normal modes and the microscopic transition probabilities leads to explicit theoretical correlations between the respective experimental quantities. Illustrative calculations of such correlations are presented for the decomposition reactions of N_2O and CO_2 diluted in Ar at shock wave temperatures, and are compared with experiment.

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On présente une analyse en mode normal perturbé pour le temps d'induction (ou d'incubation), le taux de relaxation et la vitesse de réaction d'un système unimoléculaire dilué. A température élevée, la vitesse unimoléculaire approche le comportement de Lindemann et la vitesse à basse pression est reliée aux modes normaux de relaxation des états réactifs d'une manière simple. Dans un système modèle à échelons, la relation de réseau entre les modes normaux et les probabilités de transition microscopique conduit à des corrélations théoriques explicites entre les quantités expérimentales respectives. On présente des calculs illustrant de telles corrélations pour les réactions de décomposition du N_2O et du CO_2 dilués dans Ar à des températures d'onde de choc et on les compare avec les résultats expérimentaux.

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I. Introduction

A master-equation formulation is required for a realistic treatment of thermal unimolecular reactions in the low and fall-off pressure regions (1–3). At the low pressure limit where collisional energy transfer is rate-determining, analytic solutions of the master equation have been obtained (2, 4) for several collisional transition probability models, and the possibility of extracting information on collisional energy transfer from these solutions has been the subject of much discussion (2, 5). In the fall-off pressure region where the intermolecular energy transfer and intramolecular reaction rates are comparable, the master equation is less amenable to analytic solutions, and little attention has so far been directed to the implications of the master-equation solution in this pressure region for collisional energy transfer studies.

Using the perturbed normal-mode approach (6–9), we have recently presented a master-equation formulation of thermal unimolecular reactions (3): the microscopic reaction processes were treated explicitly as dissipative perturbations to the normal modes of

collisional relaxation of the reactant; and the solution for the unimolecular reaction rate was derived analytically at all pressures in terms of the relevant microscopic reaction probabilities and relaxation elements (8). The solution relates the various experimental quantities (the unimolecular reaction rate, the relaxation rate, etc.) directly to the collisional transition probability functions of the reactant system. Where sufficient experimental data are available, it is possible to undertake a semi-quantitative analysis of the experimental data using the analytic properties of the solution, and to study the energy transfer properties of the reactant molecule. Such analyses are useful in small polyatomic systems where reliable theoretical information on collisional energy transfer properties in the reactive energy range is virtually non-existent (10). The aim of this paper is to present such analyses for the decomposition reactions of N_2O and CO_2 diluted in Ar at high temperature.

II. The Relaxation and Unimolecular Reaction Rates

Following earlier works (3, 9), we envisage the reactant molecule to possess a set of n discrete states, and define $\eta_i(t)$ as the population of state i at time t , $\tilde{n}_i$ the equilibrium population of state i in the absence

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of reaction, k_i the microscopic reaction probability of state i , and q_{ij} the transition rate constant from state i to state j via collisions with heat-bath molecules M . The master equation for the evolution of the population vector $\eta(t)$ is

$$\begin{aligned} [1] \quad d\eta(t)/dt &= \mathbf{J} \cdot \eta(t) \\ &= (\mathbf{A} + \mathbf{D}) \cdot \eta(t) \end{aligned}$$

where $a_{ij} = q_{ji}[M]$, $a_{ii} = -\sum_{j \neq i} a_{ji}$, and $d_{ij} = \delta_{ij}d_i = -\delta_{ij}k_i$. The evolution eq. [1] may be transformed (3) to

$$[2] \quad d\chi(t)/dt = \mathbf{R} \cdot \chi(t)$$

where

$$[3] \quad \chi(t) = \mathbf{E}^{-1/2} \cdot \eta(t)$$

and

$$\begin{aligned} [4] \quad \mathbf{R} &= \mathbf{E}^{-1/2} \cdot \mathbf{J} \cdot \mathbf{E}^{1/2} \\ &= \mathbf{B} + \mathbf{D} \end{aligned}$$

with

$$[5] \quad \mathbf{B} = \mathbf{E}^{-1/2} \cdot \mathbf{A} \cdot \mathbf{E}^{1/2}$$

and $e_{ij} = \tilde{n}_i \delta_{ij}$. Both $\mathbf{B}$ and $\mathbf{R}$ are symmetric, and may be diagonalised (3) to

$$[6] \quad \mathbf{B} = \mathbf{S} \cdot \mathbf{\Lambda} \cdot \mathbf{S}^t$$

and

$$[7] \quad \mathbf{R} = \mathbf{\Psi} \cdot \mathbf{\Gamma} \cdot \mathbf{\Psi}^t$$

with the respective eigenvalues λ_j and γ_j being non-positive, as follows

$$[8] \quad 0 = \lambda_{n-1} > \lambda_{n-2} \geq \lambda_{n-3} \geq \dots \geq \lambda_0$$

and

$$[9] \quad 0 > \gamma_{n-1} \gg \gamma_{n-2} \geq \gamma_{n-3} \geq \dots \geq \gamma_0$$

Formally, the solution for the population evolution eq. [1] is

$$[10] \quad \eta(t) = \mathbf{E}^{1/2} \cdot \mathbf{\Psi} \cdot \exp(\mathbf{\Gamma}t) \cdot \mathbf{\Psi}^t \cdot \mathbf{E}^{-1/2} \cdot \eta(0)$$

or equivalently,

$$[10a] \quad \eta_i(t) = \tilde{n}_i^{1/2} \sum_j \exp(\gamma_j t) \psi_{ij} \sum_k \psi_{kj} \eta_k(0) \tilde{n}_k^{-1/2}$$

At short time ($t \ll |\gamma_{n-1}|^{-1}$), the system goes through a transient phase during which the initial distribution of the reactant molecules relaxes to a new distribution via collision (VR-T) energy transfer with the heat-bath molecules; it is convenient to treat this "vibrational relaxation" phase using the approach of normal modes of relaxation (refs. 11, 12; see, however, ref. 4b for the implications at very high temperature), and relate the experimental energy

relaxation rate to the characteristic time constant of the slowest relaxation mode. Explicitly (12),

$$[11] \quad k_{\text{rel}} = \lim_{t \rightarrow \infty} -d \ln [E(t) - \tilde{E}] / dt \simeq -\lambda_{n-2}$$

$E(t)$ and $\tilde{E}$ being the average internal energy of the reactant at time t and at equilibrium, respectively.

At long time ($t > |\gamma_{n-1}|^{-1}$), the term associated with γ_{n-1} in the j -summation of eq. [10a] will dominate, and the total population of the reactant molecule becomes

$$\begin{aligned} [12] \quad \eta(t) &= \sum_i \eta_i(t) \rightarrow \left(\sum_i \tilde{n}_i^{1/2} \psi_{i,n-1} \right) \\ &\times \left[\sum_k \psi_{k,n-1} \eta_k(0) \tilde{n}_k^{1/2} \right] \exp(\gamma_{n-1} t) \end{aligned}$$

This compares with

$$[13] \quad \eta(t) = \eta(0) \exp[-k_{\text{uni}}(t - \tau_{\text{ind}})]$$

whence the unimolecular reaction rate and the induction (or incubation) time are respectively

$$[14] \quad k_{\text{uni}} = -\gamma_{n-1}$$

and

$$\begin{aligned} [15] \quad \tau_{\text{ind}} &= -\gamma_{n-1}^{-1} \left[\ln \left(\sum_i \tilde{n}_i^{1/2} \psi_{i,n-1} \right) \right. \\ &\quad \left. + \ln \left(\sum_i \psi_{i,n-1} \eta_i(0) \tilde{n}_i^{-1/2} / \eta(0) \right) \right] \end{aligned}$$

The induction time τ_{ind} comprises two parts: the first depends only upon the transition probabilities at the temperature of the reaction; the second, as shown by Brau, Keck, and Carrier (13), depends on the initial population also.

The solution of the master equation predicts (9) that, at least for weak-collision systems, the unimolecular rate expression approaches the simple Lindemann form at high temperature; viz.

$$[16] \quad k_{\text{uni}} = k_{\infty} / \{1 + k_{\infty}/k_0[M]\}$$

where k_{∞} and k_0 are respectively the infinite-pressure unimolecular and low-pressure bimolecular rate constants. This prediction has indeed been confirmed for the decomposition reactions of diluted N_2O , CO_2 , NO_2 , and N_2H_4 at shock-wave temperatures (9).

The conformity of a reaction to the Lindemann behaviour has the interesting implication that the bimolecular rate constant k_0 is related to the relaxation elements associated with the reactive states (9), which, for a step-ladder model system ($q_{ij} = 0$ for $|i-j| > 1$), are related to the detailed transition rate constants $q_{i,i+1}$ in a network-like manner (7, 8). Explicitly,

$$[17] \quad k_0 = \left(\sum_{j=0}^{n-2} t_j \right)^{-1}$$

where

$$t_j = \left(\sum_{i=0}^j \tilde{n}_i \right)^2 / \tilde{n}_j q_{j,j+1}$$

Notice that the bimolecular rate constant k_0 is independent of the microscopic reaction probabilities k_i , since collisional energy transfer is rate-determining at the low pressure limit.

III. Calculations

Under these conditions, the overall reaction rate depends principally on the average energy transfer $\langle \Delta E \rangle$ and is only weakly dependent on the detailed functional form of the collisional transition probabilities (2, 5, 7, 8, 14); moreover, it has been shown (5, 8), particularly when the relevant energy states are grouped into pseudo-levels in the actual calculation, that the step-ladder model gives a reasonable (but coarsely averaged) description of the collisional energy-transfer properties of the reactant system. Hence, it is a simple matter to calculate the bimolecular rate constant k_0 using eq. [17] given the transition rate constants $q_{i,i+1}$; also, once k_0 is determined, the eigenvector Ψ_{n-1} and hence the induction time τ_{ind} in eq. [15] may be calculated non-iteratively. Our method of calculation is then to (i) grain the energy states of the reactant molecule into pseudo-levels; (ii) characterise the transition rate constants in the form of

$$[18] \quad q_{i,i+1} = q_{01} f_i$$

with q_{01} so chosen that the relaxation eigenvalue λ_{n-2} coincides with (minus) the experimental relaxation rate (cf. eqs. [5], [6], and [11]); and (iii) examine the dependence of the calculated bimolecular reaction rate and induction time on the respective functional forms f_i .

A. Graining of Energy States

The colossal number of vibration-rotation states in the interesting energy region ($\sim 10^7$ and $\sim 10^8$ for N_2O and CO_2 , respectively, below the bond dissociation energy) necessitates the grouping of near-degenerate states into pseudo-levels — an approximation justified by the relatively large collisional energy transfer rates between levels of similar energies. The grained system then contains $\bar{n}$ pseudo-levels, with energies $\bar{\epsilon}_i$ and degeneracies $\bar{g}_i$ ($i = 0, 1, 2, \dots, \bar{n} - 1$) where

$$[19] \quad \bar{\epsilon}_i = i \bar{\epsilon}$$

and

$$[20] \quad \bar{g}_i = G\left(\frac{2i+1}{2} \bar{\epsilon}\right) - G\left(\frac{2i-1}{2} \bar{\epsilon}\right)$$

the sum of states $G(\epsilon)$ up to energy ϵ being computed by direct count (15) for $\epsilon \leq 4000 \text{ cm}^{-1}$ and by steepest-descent (16) for $\epsilon > 4000 \text{ cm}^{-1}$. Notice that the graining scheme covers energy states from $\epsilon_{\text{min}} = 0$ to $\epsilon_{\text{max}} = (\bar{n} + \frac{1}{2})\bar{\epsilon}$.

The state sum $G(\epsilon)$ depends upon the chosen model for the reactant molecule. Both in diatomic and in polyatomic systems, rotation has often been ignored altogether in the enumeration of the state-sum. Yet, as we have shown (8), rotational effects are important in diatomic dissociation; rotational dissociation is believed to be important in polyatomic molecules also. In addition, there are no arbitrary reasons to exclude rotation in collisional energy transfer. Hence, we examine the effect of inclusion of rotation in the state-sum using two series of calculations: one including both vibration and rotation (VR) in the state-count and the other including vibration only (V), with the vibrations treated as independent oscillators in both cases.

In N_2O , the two stretching modes are well represented by two independent Morse oscillators (17); the two degenerate bending modes are treated as harmonic. For CO_2 , all four vibrations are treated as harmonic. The relevant spectroscopic parameters are listed in Table 1.

The grain size $\bar{\epsilon}$ is chosen to be one vibrational quantum for the bending mode (600 and 700 cm^{-1} for N_2O and CO_2 , respectively); other grain sizes have also been used in test calculations to examine the sensitivities of the calculated quantities with variations in $\bar{\epsilon}$, see below. The topmost pseudo-level, $\bar{n}$, is chosen so that $\bar{\epsilon}_{\bar{n}}$ lies above the critical energy of reaction, ϵ_{crit} . The latter quantity is actually uncertain; it corresponds to the crossing point of the singlet and triplet adiabatic potentials, and is placed at ~ 64 and $\sim 125 \text{ kcal/mol}$ for N_2O and CO_2 respectively, by experiments (18, 19) and by theoretical calculations (20). Thus, we set $\bar{n} = 39$ for N_2O , corresponding to $\bar{\epsilon}_{\bar{n}-1} = 63.5$ and $\bar{\epsilon}_{\bar{n}} = 65.2 \text{ kcal/mol}$. (For CO_2 , $\bar{n} = 64$, $\bar{\epsilon}_{\bar{n}-1} = 124.1$, and $\bar{\epsilon}_{\bar{n}} = 126.1 \text{ kcal/mol}$.) Again, the value of $\bar{\epsilon}_{\bar{n}}$ was varied in test calculations; the calculated rates are not sensitive to the uncertainties in $\bar{\epsilon}_{\bar{n}}$ since they depend primarily on the transition probabilities associated with levels below the reactive energy region.

B. Transition Probability Models

Within the framework of the step-ladder model, two functional forms of f_i (cf. eq. [18]) were examined: the constant f_i model (CF), with

TABLE 1. Spectroscopic parameters in reactant molecule

Molecule	$\omega_e(x_e)^a$				B_x (cm ⁻¹)	Reference
N ₂ O	1298.4 (0.0048)	599.4	599.4	2283.1 (0.0066)	0.41901	35, 36
CO ₂	1388.2	667.4	667.4	2349.2	0.39021	36

$$^a \omega_e = \omega_e [(1 - x_e)v - v^2 x_e], \omega_e \text{ in cm}^{-1}.$$

$$[21a] \quad q_{i,i+1} = q_{01}$$

for all i ; and the Landau-Teller model (LT), with

$$[21b] \quad q_{i,i+1} = (i + 1)q_{01}$$

For each model, q_{01} is related through eqs. [5], [6], and [11] to the experimental relaxation time,² the latter being given by

$$[22] \quad \log_{10} p\tau_{rel} (\text{atm } \mu\text{s}) = -1.36 + 13.6T^{-1/3} \\ (800 < T < 2370 \text{ K})$$

for N₂O in Ar (21) and

$$[23] \quad \log_{10} p\tau_{rel} (\text{atm } \mu\text{s}) = -1.56 + 21.3T^{-1/3} \\ (450 < T < 3000 \text{ K})$$

for CO₂ in Ar (22, 23).

C. The Reaction Rate

It should be reiterated that eq. [17] is valid only when the unimolecular fall-off is Lindemann-like, i.e., when eq. [16] is obeyed. Hence, we shall restrict our calculations in this Section to $T > 2000$ K for N₂O and $T > 3000$ K for CO₂, where the experimental rates have been found to conform to eq. [16] (9).

N₂O

The low-pressure dissociation rates of N₂O diluted in Ar have been measured by several authors. The high temperature data (2160–3590 K) of Dove *et al.* (21) are in good agreement with those of Olschewski *et al.* (25) at lower temperatures (1400–2500 K); the data of Gutman *et al.* (26) are in good agreement with both Dove *et al.* (21) and Olschewski *et al.* (25) near 2000 K, but are systematically lower at higher temperatures. Figure 1 compares these experiments with three calculations. The first one, labelled VR-LT, includes rotation in the state-count and assumes the LT transition probability model for the transition rates between the *grained* levels (eq. [21b]).

²The expression $q_{01} = k_{rel}/[\exp((\epsilon_1 - \epsilon_0)/kT) - 1]$, which is used extensively (8, 24) to relate q_{01} to the experimental relaxation rate in diatomic systems, is not applicable to polyatomic systems. Hence, we compute the eigenvalue λ_{n-2}' for the reduced transition matrix A' (A' being constructed with $q_{i,i+1}' = f_i$) and relate q_{01} to the experimental relaxation rate using the relationship $q_{01} = -k_{rel}/\lambda_{n-2}'$. For the transition probability patterns that we examined, this latter expression can differ from the former one by as much as a factor of ten.

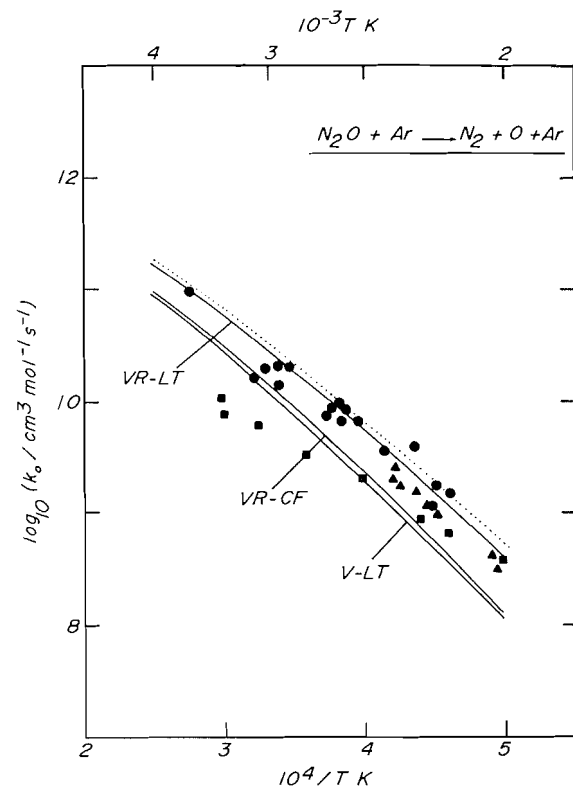


FIG. 1. Low-pressure rate constant for dissociation of N₂O diluted in Ar. Experiments: ●, Dove *et al.* (21); ▲, Olschewski *et al.* (25) with 0.2% N₂O in argon; ■, representative data points of Gutman *et al.* (26). Calculations — solid curves: VR-LT, vibration-rotational state-count and Landau-Teller transition probability model, eq. [21b]; VR-CF, vibration-rotational state-count and constant f_i model, eq. [21a]; V-LT, vibrational state-count, and Landau-Teller model. Calculations — dotted curve: VR-LT calculation with a step-size of 1200 cm⁻¹ in LT model (see text).

The second, labelled VR-CF, differs from the first one in that it assumes the constant f_i model (eq. [21a]) instead. The third calculation, labelled V-LT, excludes rotation in the state-count and assumes the LT model. Notice that the absolute magnitude of the transition probabilities in the three models are chosen to yield the same (experimental) relaxation rate (see previous subsection). Yet, they predict appreciably different reaction rates. This is not unexpected since relaxation and reaction rates are in general related to the collisional energy transfer

properties of the reactant molecule at different energy regions. It is now well recognized (14) that they are the consequences of a network of microscopic processes, and it is generally not possible to associate them with a particular transition (27). The energy region of importance to the reaction rate is characterised by the bottleneck (7, 8), which shifts to lower energy as temperature increases. The corresponding region for the relaxation rate is associated with levels carrying the largest energy fluxes; it is more subtle and a normal-mode analysis is needed for a detailed understanding (11).

That the VR-CF rates are smaller than the experimental rates (of Dove *et al.* (21) and Olschewski *et al.* (25)) by a factor of two to three is not surprising, since the transition rates are expected to increase at higher energy, rather than being constant at all energies. Hence, eq. [21a] represents an underestimate for the transition rates in the reactive energy region; and this underestimate is reflected in the calculated reaction rates.

Likewise, the V-LT rates are expected to be smaller than the experimental data, since the exclusion of rotation in the state-count underestimates the state-density (and population) of the reactive energy levels.

Note, however, that the VR calculations are likely to be (slight) overestimates, since in the graining procedure, we have grouped all energy states above the critical energy for reaction into reactive pseudo-levels, whereas in reality some of these states (those in very high rotational states) are non-reactive. We believe it is this overestimate that compensates for the underestimate in the LT transition rate constants, yielding the good agreement between the VR-LT rates and experiment; while the present calculations suggest the LT model to be much more realistic than the CF model, we would hesitate to attach any physical interpretation or justification to the model.

The activation energies of the three calculated rates are compared in Table 2. (i) The inclusion of rotation in the state-count lowers the activation energy (difference between VR-LT and V-LT) by 5 kcal/mol at 2100 K and more at higher temperatures; the reasons for this lowering have been discussed earlier in diatomic dissociation (8). (ii) The lower activation energies of the VR-LT rates, compared with the corresponding VR-CF data, arise from the faster energy transfer rates at high energy in the LT model and hence more drastic population disequilibrium at high temperature: the reasons for this have also been discussed before (8).

To examine the effect of the grain size $\bar{\epsilon}$ on the calculated reaction rate, the VR-LT calculation was

TABLE 2. Activation energy for the dissociation of N_2O in Ar at low pressure

T_1	T_2	$\epsilon_{act,0}^a$		
		VR-LT	V-LT	VR-CF
4000	3500	42.1	48.0	45.4
3500	3250	43.5	49.6	47.4
3250	3000	44.9	50.9	49.2
3000	2750	46.5	52.5	51.2
2750	2500	48.5	54.3	53.5
2500	2300	50.5	56.0	55.7
2300	2100	52.4	57.5	57.6

$$^a \epsilon_{act,0} = -k \ln [k_0(T_1)/k_0(T_2)]/[1/T_1 - 1/T_2], \text{ in kcal/mol.}$$

repeated using $\bar{\epsilon} = 1200 \text{ cm}^{-1}$ (instead of 600 cm^{-1}); the calculated rates increase by less than 30% (dotted curve in Fig. 1), showing that the calculation is indeed not sensitive to the grain size chosen.

CO_2

The dissociation of diluted CO_2 at shock wave temperatures has received much experimental attention in the last two decades; nevertheless, data of various workers were not always concordant, see ref. 28. The experimental activation energy at low pressure is significantly lower than the bond dissociation energy ($\epsilon_{act,0} \sim 105 \text{ kcal/mol}$ at 3300 K (29) compared with $D(O-CO)$: 125.75 kcal/mol). Theoretically, Gilbert (30) obtained from model calculations an activation energy of 98 kcal/mol at 4000 K. However, his calculations assume energy-independent collision cross sections, and it is not clear if these cross sections are in fact consistent with the temperature dependence of the relaxation rate. Since the activation energy for reaction is related to that of the relaxation rate, it is not possible to assess the calculated activation energies with respect to experiment. More recently, Tardy (31) obtained $\epsilon_{act,0} \simeq 100 \text{ kcal/mol}$ at 3500 K by applying corrections to the strong-collision rate constants and using his "universal function" for collision efficiency. Again, the consistency between the experimental relaxation rate and the transition probabilities assumed in this calculation is not at all clear. Both of these calculations ignore rotation; inclusion of rotation would lower the calculated activation energies substantially.

In this section, we calculate the reaction rate by synthesising the transition probabilities from eq. [21] (a and b) and experimental relaxation rate data. Figure 2 compares our calculations with the experimental data of Dean (32), Vasatko (33), Davies (34), and Wagner and Zabel (29). The data of Dean and Vasatko are in good agreement; those of Wagner and Zabel (29) are systematically lower. Again, the

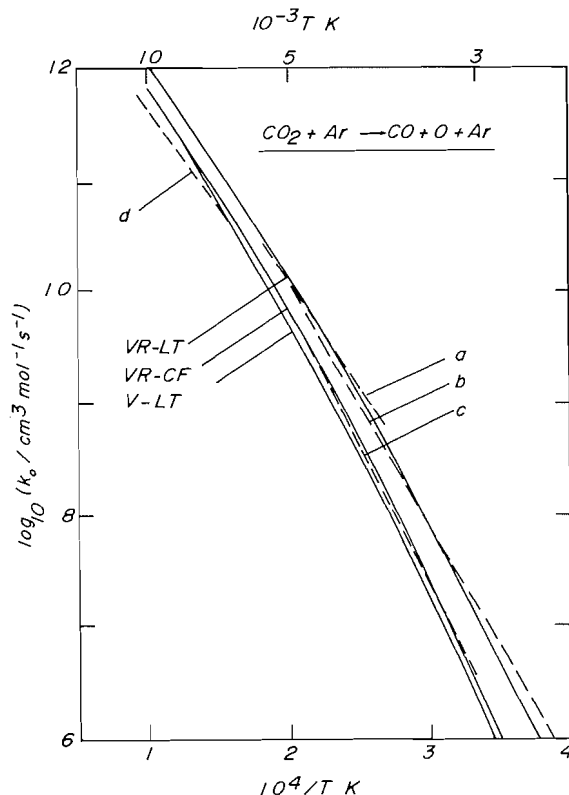


FIG. 2. Low-pressure rate constant for dissociation of CO_2 diluted in Ar. Experiments: *a*, Dean (32); *b*, Vasatko (33); *c*, Wagner and Zabel (29); *d*, Davies (34). Calculations: see caption in Fig. 1 for details.

VR-LT rates, which are in good agreement with the experiments of Dean (32) and Vasatko (33), are a factor of 2–4 larger than the V-LT and VR-CF rates.

Table 3 compares the activation energies of the respective rates. (i) Again, rotation leads to a noticeable reduction in the activation energy: from 7 kcal/mol at 3500 K to 10 kcal/mol at 10 000 K. (ii) Note also the dependence of the activation energy on the transition probability model, which becomes more appreciable at high temperature; cf. the VR-LT vs. VR-CF data. (iii) The activation energies of the VR-LT rates are in good accord with experimental estimates: the calculated value of $\epsilon_{\text{act},0} = 109.3$ kcal/mol between 3000 and 3500 K compares favourably with the experimental value (29) of 105 kcal/mol at 3300 K. Clearly, the low activation energies at very high temperatures are not anomalous at all.

D. The Induction Time

In the relaxation experiment of Dove *et al.* (21), the induction time for the N_2O reaction was measured and reported in terms of its ratio to the vibrational relaxation time, $\tau_{\text{ind}}/\tau_{\text{rel}}$. The ratio ranges

TABLE 3. Activation energy for the dissociation of CO_2 in Ar at low pressure

T_1	T_2	$\epsilon_{\text{act},0}^a$		
		VR-LT	V-LT	VR-CF
10000	7500	79.1	89.4	82.9
7500	6000	83.2	94.6	89.2
6000	5000	89.8	101.2	97.3
5000	4500	98.1	108.2	106.6
4000	3500	104.9	113.2	113.4
3500	3000	109.3	116.3	117.4

^aSee footnote in Table 2.

TABLE 4. Induction to relaxation time ratio of the N_2O -Ar system

T (K)	$\tau_{\text{ind}}/\tau_{\text{rel}}$		
	VR-LT	V-LT	VR-CF
4000	2.13	2.05	1.47
3000	2.37	2.31	1.78
2500	2.56	2.48	2.10
2100	2.72	2.58	2.58

TABLE 5. Induction to relaxation time ratio of the CO_2 -Ar system

T (K)	$\tau_{\text{ind}}/\tau_{\text{rel}}$		
	VR-LT	V-LT	VR-CF
10000	2.08	2.00	1.33
6000	2.42	2.43	1.73
4000	2.94	2.83	2.42
3500	3.03	2.71	1.74

between 4 and 7 at 2160–3590 K. To our knowledge, no measurements exist for the CO_2 reaction.

From eq. [15], τ_{ind} is proportional to the reaction time $\tau_{\text{react}} (= -1/\gamma_{n-1})$. Hence, it is in general more convenient to characterise the induction time in terms of the ratio $\tau_{\text{ind}}/\tau_{\text{react}}$. Nevertheless, in the low-pressure limit, τ_{react} is directly proportional to τ_{rel} , and it is equally convenient to characterise τ_{ind} in terms of $\tau_{\text{ind}}/\tau_{\text{rel}}$; to facilitate comparison with experiment, we present our results in this section in this form.

Tables 4 and 5 list the ratios $\tau_{\text{ind}}/\tau_{\text{rel}}$ for the N_2O and CO_2 reactions, respectively, for the three sets of calculations.³ Notice that τ_{ind} is dependent upon the initial population distribution, $\eta(0)$; nevertheless, as for relaxation time, the dependence is not significant

³Kiefer and Hajduk (24) have recently cautioned against the numerical difficulties associated with the use of eq. [15]. We did not encounter any difficulties in our calculations. However, the alleged difficulties may well arise in calculations at lower temperatures.

for polyatomic systems in experimental shock conditions: in all calculations, an initial temperature of 300 K was assumed to mimic the experiment. For both reactions, the ratio decreases slightly with increasing temperature, in accord with the experiment of Dove *et al.* (21); the calculated ratios of 2–3 are in reasonable agreement with the experimental ratios of 4–7, considering the simplicity of the theoretical model and the difficulty of the experiment. Note the similar patterns between the corresponding data in the two reactions: the VR-LT and V-LT induction times are very similar; on the other hand, the difference between the VR-LT and VR-CF induction times is quite appreciable, especially at high temperature. This is because the induction time is related predominantly to the transition probability pattern, and less so to the energy level structure of the reactant molecule.

IV. Discussions

The present master-equation analysis of the relaxation and reaction behaviour of the N_2O -Ar and CO_2 -Ar systems provides interesting insights into the energy transfer properties of small polyatomics ("weak collision systems") at high temperature.

In the master-equation formulation, the reaction rate at low pressure is explicitly related to the internal (collisional) relaxation behaviour of the reactant. It is important to note that the relaxation rate has, in general, no direct connection with the "strong" collision rate. (Indeed, the transition rate constant q_{01} is not related to the Lennard-Jones collision number at all!) Nor is there any similar relationship between the (non-equilibrium) reaction rate and the strong collision number. Thus, to relate these last two quantities using a collisional efficiency parameter has, at best, operational meaning only.⁴ For meaningful interpretations of a reaction at high temperature and the energy transfer properties of the reactant, the collisional relaxation behaviour of the molecular system must be explicitly considered. Note also that it is in general not possible to relate the reaction rate to the collision (or energy transfer) rate from a particular energy state or group of states in the reactant molecule; rather, the overall rate represents a co-operative effect of a network of microscopic transition processes.

The treatment of the microscopic reaction processes in unimolecular systems as dissipative pertur-

bations to the internal (collisional) relaxation of the reactant molecule is attractive for several reasons. The algebra involved in the treatment bears close resemblance to perturbation formalisms in quantum mechanics; this facilitates not only the analytic solution for the unimolecular reaction rate at all pressures, but also the physical interpretations of the relevant theoretical results. Thus, the theoretical origin of the Lindemann behaviour is now well understood: it occurs *either* when the molecule has only one reactive state, in which case the internal relaxation amongst its various states has the overall effect of grouping all the non-reactive states into one (3); *or* when the relaxation modes associated with the respective reaction states (as characterised by the respective relaxation elements (8)) are identical, in which case the reaction-relaxation behaviour of all the reactive states is similar. The latter follows from the fact that the reaction bottleneck lies in the non-reactive energy region, and that the reaction behaviour does not depend at all on the microscopic processes associated with the reactive states (9).

This latter point is important since it leads to the prediction of the Lindemann fall-off behaviour at high temperature and, in turn, the direct association of the low-pressure reaction rate at high temperature with the relaxation elements of the reactive states. Within the step-ladder model, the justification of which for a grained system we have discussed earlier, this direct identification facilitates a straightforward semiquantitative analysis of the collisional energy transfer properties of a reactant system, from the experimental relaxation and reaction rate data.

The present master-equation analysis of the various experimental data (the relaxation rate, the reaction rate, and the induction time) of the N_2O and CO_2 systems demonstrates that (i) rotation should be taken into account in rate calculations; (ii) both "vibrational" population disequilibrium and rotation are important contributors to the lowering of the low pressure activation energy at high temperature; and (iii) an (initial-) state-independent energy transfer model is inadequate, even for the description of thermal systems; in the model we have used, the initial state is characterised only by its total energy (and not by its rotational and vibrational components), but within this framework of coarsely-grained pseudo-levels, a Landau-Teller transition probability model seems to give semi-quantitatively reasonable results. As we have noted before (8) it is not rotational energy transfer nor rotational disequilibrium that are of vital importance in determining the reaction rate at these temperatures, or its Arrhenius temperature coefficient: it is simply the proper accounting for the rotational states taken

⁴The "strong collision" rate constants for the two reactions have recently been calculated by Troe (2), using slightly different spectroscopic data. For N_2O , the collisional efficiency decreases from 0.09 at 900 K to 0.02 at 2500 K; for CO_2 it increases from 0.005 at 1000 K to 0.03 at 3000 K.

together with the fact that at high temperatures, rotational energy and vibrational energy seem to be roughly equally effective in promoting dissociation (8).

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